Resultant of Coplanar Non-Concurrent Force Systems

High-Level Overview

In Statics and Engineering Mechanics, a **force system** consists of multiple forces and/or moments acting simultaneously on a body. When these forces act in a single 2D plane, they are called **coplanar**. If their lines of action do not all intersect at a single point, they form a **coplanar non-concurrent force system**.

To analyze the combined effect of such a system, we reduce all forces and moments into a single equivalent force called the **Resultant Force (R)**, acting at a specific line of action (or perpendicular distance d) relative to a given reference point (such as the origin or base O).

Fundamental Terminology

- **Force Vector ($\mathbf{F}$):** A quantity having both magnitude and direction, measured in Newtons (N) or Kilonewtons (kN).
- **Moment (M):** The rotational effect produced by a force about a specific point, measured in Newton-meters ($\mathrm{N \cdot m}$) or Kilonewton-meters ($\mathrm{kN \cdot m}$).

$$\text{Moment} = \text{Force} \times \text{Perpendicular Distance}$$

- **Resultant Force (R):** A single force that produces the exact same translational effect as the combined system of original forces.
- **Varignon's Theorem (Principle of Moments):** The algebraic sum of moments produced by individual forces about any point is equal to the moment produced by their single resultant force about that same point.

Step-by-Step Analytical Breakdown

To determine the resultant force and its point/distance of application for a coplanar force system, follow this standardized procedure:

1. Vector Resolution of Forces

Decompose every inclined force acting at an angle θ into horizontal (x) and vertical (y) components:

- Horizontal component: $F_x = F \cos\theta$
- Vertical component: $F_y = F \sin\theta$

Sign Conventions:

- Rightward forces ($\rightarrow$) are positive ($+$); leftward forces ($\leftarrow$) are negative ($-$).
- Upward forces ($\uparrow$) are positive ($+$); downward forces ($\downarrow$) are negative ($-$).

2. Summation of Force Components

Calculate the algebraic sum of all horizontal and vertical forces:

$$\Sigma F_x = \text{Sum of all horizontal force components}$$

$$\Sigma F_y = \text{Sum of all vertical force components}$$

3. Magnitude and Angle of Resultant

The total magnitude R of the resultant force is found using the Pythagorean theorem:

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

The angle θ that the resultant makes with the horizontal axis is given by:

$$\theta = \tan^{-1}\left(\left|\frac{\Sigma F_y}{\Sigma F_x}\right|\right)$$

4. Moment Summation & Application of Varignon's Theorem

Choose a reference point (e.g., origin O or point A). Compute the net moment ΣM_O^F exerted by all individual forces and explicit moment couples about that point:

Sign Conventions for Moments:

- Counter-Clockwise (CCW) moment = Positive $(+)$
- Clockwise (CW) moment = Negative $(-)$

Apply Varignon's Theorem to find the perpendicular distance d from the reference point to the line of action of R:

$$\Sigma M_O^F = R \times d \implies d = \frac{|\Sigma M_O^F|}{R}$$

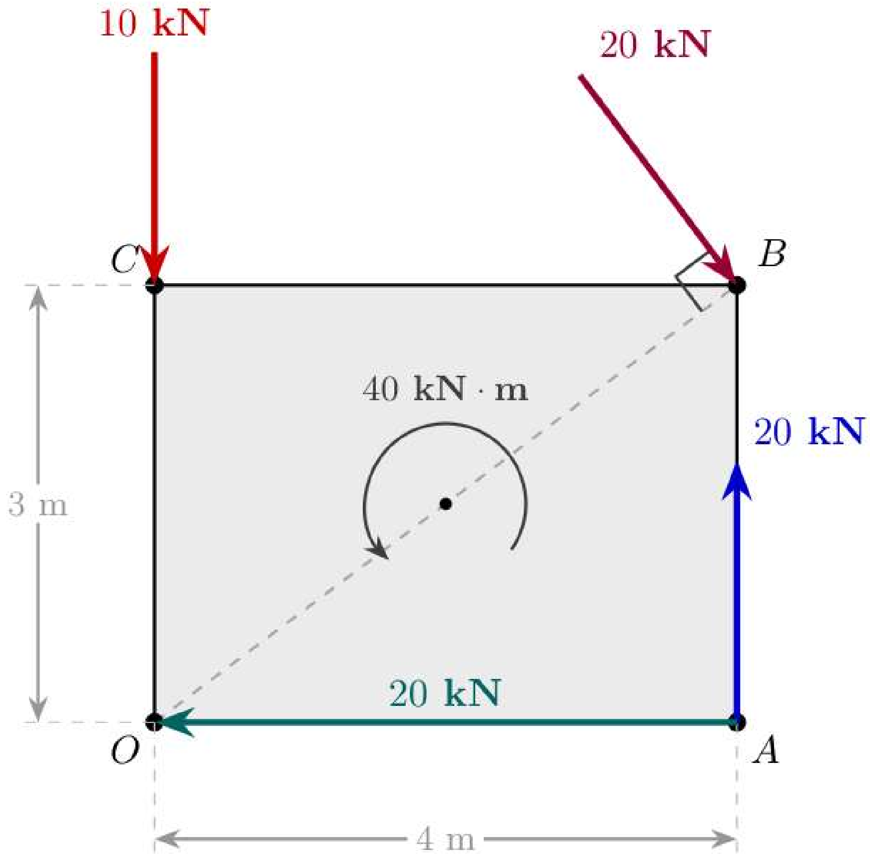
Worked Examples

Problem 1: Rectangular Body $OABC$ Under Multiple Forces and a Couple Moment

Problem Statement

Find the resultant of the force system acting on a body $OABC$ shown in the schematic. Determine the perpendicular distance of the resultant from point O.

- Dimensions: Width $OA = BC = 4\,\mathrm{m}$, Height $OC = AB = 3\,\mathrm{m}$.
- Point O is located at $(0, 0)$.
- Point A is located at $(4, 0)$.
- Point B is located at $(4, 3)$.
- Point C is located at $(0, 3)$.
- Applied Forces/Moments:
    i. A downward force of $10\,\mathrm{kN}$ at point C.
    ii. An upward force of $20\,\mathrm{kN}$ at point A.
    iii. A leftward force of $20\,\mathrm{kN}$ along segment OA at point O.
    iv. An inclined force of $20\,\mathrm{kN}$ acting inward at vertex B, perpendicular to diagonal OB.
    v. A counter-clockwise couple moment of $40\,\mathrm{kN}\cdot\mathrm{m}$ at the center.



Visual Description: A rectangular body $OABC$ in the first quadrant of a Cartesian plane. Corner O is at $(0, 0)$, A at $(4, 0)$, B at $(4, 3)$, and C at $(0, 3)$. Diagonal OB is shown as a dashed line. A $20\,\text{kN}$ vertical force acts upwards at A. A $20\,\text{kN}$ horizontal force acts leftwards along OA towards O. A $10\,\text{kN}$ vertical force acts downwards at C. A $20\,\text{kN}$ force acts at B, oriented perpendicular to diagonal OB. A counter-clockwise arc represents a $40\,\text{kN} \cdot \text{m}$ moment at the center.

Detailed Solution

Step 1: Geometric Analysis of Diagonal OB

The angle θ_1 made by diagonal OB with the horizontal line OA is:

$$\tan\theta_1 = \frac{\text{Height}}{\text{Width}} = \frac{3}{4}$$

$$\theta_1 = \tan^{-1}\left(\frac{3}{4}\right) = 36.869^\circ$$

Since the $20\,\text{kN}$ force applied at point B is perpendicular to diagonal OB, the angle θ_2 that this force makes with the horizontal line is:

$$\theta_2 = 90^\circ - \theta_1 = 90^\circ - 36.869^\circ = 53.13^\circ$$

Resolving the $20\,\text{kN}$ force at point B:

- Horizontal component (rightward): $20\cos(53.13^\circ) \approx 12\,\text{kN}$
- Vertical component (downward): $20\sin(53.13^\circ) \approx 16\,\text{kN}$

Step 2: Summation of Forces

Summing horizontal forces along the x-axis:

$$\Sigma F_x = -20 + 20\cos(53.13^\circ)$$

$$\Sigma F_x = -20 + 12 = -8\,\text{kN}$$

Therefore, $\Sigma F_x = 8\,\text{kN} \quad (\leftarrow)$

Summing vertical forces along the y-axis:

$$\Sigma F_y = 20 - 10 - 20\sin(53.13^\circ)$$

$$\Sigma F_y = 20 - 10 - 16 = -6\,\text{kN}$$

Therefore, $\Sigma F_y = 6\,\text{kN} \quad (\downarrow)$

Step 3: Resultant Magnitude and Direction

Magnitude:

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2} = \sqrt{(-8)^2 + (-6)^2} = \sqrt{64 + 36} = \sqrt{100} = 10\,\text{kN}$$

Direction relative to the horizontal:

$$\theta = \tan^{-1}\left(\left|\frac{\Sigma F_y}{\Sigma F_x}\right|\right) = \tan^{-1}\left(\frac{6}{8}\right) = 36.869^\circ$$

The resultant R acts at an angle of 36.869° pointing into the third quadrant (downward and leftward).

Step 4: Net Moment About Point O

Taking counter-clockwise moments as positive $(+)$:

1. $20\,\text{kN}$ force at A (upward): Distance $= 4\,\text{m}$, Moment $= +(20 \times 4) = +80\,\text{kN}\cdot\text{m}$ (CCW)
2. $10\,\text{kN}$ force at C (downward): Passes through vertical line OC ($d = 0\,\text{m}$), Moment $= 0$
3. $20\,\text{kN}$ force at O (leftward): Line of action passes directly through point O, Moment $= 0$
4. Components of $20\,\text{kN}$ force at B:
   - Horizontal component $20\cos(53.13^\circ)$ (rightward) at height $y = 3\,\text{m}$: Moment $= -(20\cos(53.13^\circ) \times 3) = -(12 \times 3) = -36\,\text{kN}\cdot\text{m}$ (CW)
   - Vertical component $20\sin(53.13^\circ)$ (downward) at distance $x = 4\,\text{m}$: Moment $= -(20\sin(53.13^\circ) \times 4) = -(16 \times 4) = -64\,\text{kN}\cdot\text{m}$ (CW)
5. Pure couple moment applied at center: $+40\,\text{kN}\cdot\text{m}$ (CCW)

Summing all moments about point O:

$$\Sigma M_O^F = +80 - 36 - 64 + 40 = 20\,\text{kN}\cdot\text{m} \quad (\text{Counter-Clockwise})$$

Step 5: Perpendicular Distance from Point O

Using Varignon's Theorem:

$$\Sigma M_O^F = R \times d$$

$$20 = 10 \times d \implies d = 2\,\mathrm{m}$$

The line of action of the resultant force $R = 10\,\mathrm{kN}$ is located at a perpendicular distance $d = 2\,\mathrm{m}$ from origin O.

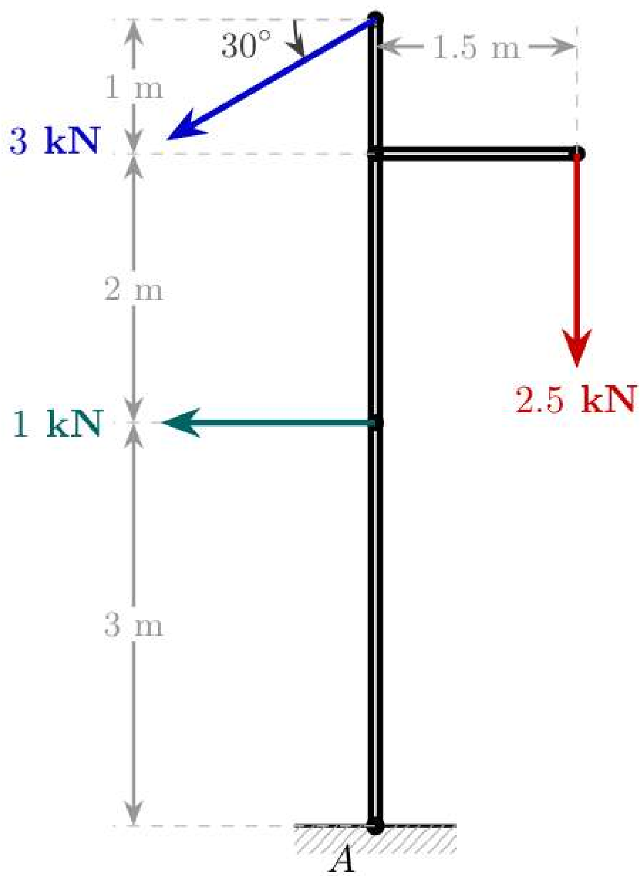
Problem 2: Vertical Frame/Pole Subjected to Inclined and Point Forces

Problem Statement

Three forces act on a vertical pole of total height $6\,\mathrm{m}$ fixed at base A:

1. A horizontal force of $1\,\mathrm{kN}$ acting leftward at height $y = 3\,\mathrm{m}$.
2. An inclined force of $3\,\mathrm{kN}$ acting at the top of the pole ($y = 6\,\mathrm{m}$) at an angle of 30° below the horizontal pointing downward-leftward.
3. A downward vertical force of $2.5\,\mathrm{kN}$ acting at the tip of a horizontal bracket extending $1.5\,\mathrm{m}$ to the right, attached at height $y = 5\,\mathrm{m}$.

Find the magnitude, direction, and line of action of the resultant force with respect to base A.



> **Visual Description:** A vertical pole attached to a fixed base at A. At top height $6\,\mathrm{m}$, an inclined $3\,\mathrm{kN}$ force acts downwards and leftwards at 30° below horizontal. A horizontal cantilever bracket extends $1.5\,\mathrm{m}$ rightwards at height $5\,\mathrm{m}$, carrying a $2.5\,\mathrm{kN}$ vertical downward force at its end. A horizontal leftward force of $1\,\mathrm{kN}$ acts directly on the main vertical pole at height $3\,\mathrm{m}$.

Detailed Solution

Step 1: Resolving Inclined Forces

The $3\,\mathrm{kN}$ force at the top ($y = 6\,\mathrm{m}$) is resolved into:

- Horizontal component (leftward): $3\cos(30^\circ) \approx 2.598\,\mathrm{kN}$

- Vertical component (downward): $3\sin(30^\circ) = 1.5\,\text{kN}$

Step 2: Summation of Forces

Sum of horizontal forces (ΣF_x):

$$\Sigma F_x = -1 - 3\cos(30^\circ) = -1 - 2.598 = -3.598\,\text{kN}$$

Therefore, $\Sigma F_x = 3.598\,\text{kN} \quad (\leftarrow)$

Sum of vertical forces (ΣF_y):

$$\Sigma F_y = -2.5 - 3\sin(30^\circ) = -2.5 - 1.5 = -4.000\,\text{kN}$$

Therefore, $\Sigma F_y = 4\,\text{kN} \quad (\downarrow)$

Step 3: Resultant Magnitude and Direction

Magnitude:

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2} = \sqrt{(-3.598)^2 + (-4)^2} = \sqrt{12.946 + 16} = \sqrt{28.946} \approx 5.38\,\text{kN}$$

Orientation angle θ:

$$\theta = \tan^{-1}\left(\frac{|\Sigma F_y|}{|\Sigma F_x|}\right) = \tan^{-1}\left(\frac{4}{3.598}\right) \approx 48.03^\circ \quad \text{(below horizontal)}$$

Step 4: Moment Summation About Base Point A

1. $1\,\text{kN}$ horizontal force at $y = 3\,\text{m}$ (leftward):

$$\text{Moment} = +(1 \times 3) = +3\,\text{kN}\cdot\text{m} \quad \text{(CCW)}$$

2. $2.5\,\text{kN}$ vertical force at horizontal distance $x = 1.5\,\text{m}$ (downward):

$$\text{Moment} = -(2.5 \times 1.5) = -3.75\,\text{kN}\cdot\text{m} \quad \text{(CW)}$$

3. $3\,\text{kN}$ inclined force components at $y = 6\,\text{m}$:
   - Horizontal component $3\cos(30^\circ)$ (leftward) at height $y = 6\,\text{m}$:

$$\text{Moment} = +(3\cos(30^\circ) \times 6) = +(2.598 \times 6) = +15.588\,\text{kN}\cdot\text{m} \quad \text{(CCW)}$$

   - Vertical component $3\sin(30^\circ)$ acts along the centerline of the pole ($x = 0\,\text{m}$):

$$\text{Moment} = 0$$

Net Moment about A:

$$\Sigma M_A^F = +3 - 3.75 + 15.588 = +14.838\,\text{kN}\cdot\text{m} \quad \text{(Counter-Clockwise)}$$

Step 5: Perpendicular Distance from A

$$d = \frac{|\Sigma M_A^F|}{R} = \frac{14.838}{5.38} \approx 2.76\,\text{m}$$

Summary Equations Reference Table

Parameter	Formula	Description
Horizontal Force Sum	$\Sigma F_x = \sum F_i \cos\theta_i$	Vector sum of all x-components

Parameter	Formula	Description
Vertical Force Sum	$\Sigma F_y = \sum F_i \sin\theta_i$	Vector sum of all y-components
Resultant Magnitude	$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$	Magnitude of net combined force
Resultant Angle	$\theta = \tan^{-1}\left\lvert\frac{\Sigma F_y}{\Sigma F_x}\right\rvert$	Angle relative to horizontal axis
Varignon's Principle	$\Sigma M_O = R \times d$	Equates net component moments to total resultant moment
Line of Action Offset	$d = \frac{\lvert\Sigma M_O\rvert}{R}$	Perpendicular distance from reference point